

Yubin Kim

E: yubink.cs@gmail.com / P: 412-204-7134

HIGHLIGHTS

- 4+ yr in ML leadership
- Total 11+ yr in ML
- Pragmatic applied ML for business impact
- Strong ML/NLP/IR research background
- Solves large-scale healthcare and e-commerce problems
- Prioritizes empathy and collaboration

SKILLS

Deep Learning & ML

- Graph Neural Network
- Transformer/LLM (e.g. BERT, GPT)
- RNN, LSTM
- Clustering

Search/Recommendations

- Evaluation
- A/B experimentation
- Efficiency
- Learning-to-rank
- Distributed search
- Knowledge graphs

Tools and Frameworks

- Python, Java, C++, SQL, Rust
- Pytorch, Scikit-learn
- Lucene, Solr, Elasticsearch
- Kubeflow, Airflow, Beam
- AWS, GCP
- Jupyter, vim, git

PERSONAL

- West coast swing dancer & competitor: *ask me about my Jack & Jill results*
- Enthusiastic gardener: *ask me about my volunteer pumpkins*

Applied AI/ML leader with 4+ yrs of leadership experience building collaborative, multi-disciplinary teams to pragmatically solve problems at scale in large e-commerce and healthcare organizations for business impact. Total 11+ yrs in machine learning, natural language processing, information retrieval, recommendations. Strong research background with publications in top tier conferences and active ties to academia.

EDUCATION

- Ph.D. in Computer Science from the Language Technologies Institute (Dec 2018)
Carnegie Mellon University, Pittsburgh, Pennsylvania
- Bachelor of Software Engineering (Apr 2011)
University of Waterloo, Waterloo, Ontario

PROFESSIONAL EXPERIENCE

- Senior Engineering Manager at **Etsy, Inc.** (04/2022–10/2023)
 - Built real time, large scale retrieval systems powering search, recs, and ads at etsy.com, an e-commerce marketplace with 90M+ active buyers and 100M items
 - Led product ML team (~5) of Intermediate–Sr. Staff Applied Science and Eng ICs in the Search Retrieval initiative of the Search org, providing technical mentorship, managing hiring, performance, team processes
 - Directly generated \$40M+ in GMS over 1.5 years by testing and deploying innovations in graph ML models and Solr through launching A/B experiments
 - Enabled \$30M+ in revenue through collaborations with the Ads org
 - Co-authored the 3-year technical strategy for the Search Retrieval initiative
 - Initiated key collaborations with analytics & product partners and cross-org enablement teams to develop new analytic metrics and tooling for the Retrieval initiative; increased the speed of online experimentation by 2x
 - Developed new metrics and processes for offline evaluation; increased successful online experiments in the Retrieval initiative by 2x within 2 quarters
- Director of Technology at **UPMC Enterprises** (07/2019–03/2022)
 - Led product team (~12) of Eng/QA ICs, Product/Eng Managers; built an Elasticsearch-based search engine over 160M+ clinical documents
 - Led applied ML team (~3) of ML engineers, responsible for creating career paths, hiring, performance management, technical mentorship
 - Collaborated with product and provider/payor stakeholders to solve problems in claims processing, automated coding, risk stratification, retrospective analysis
 - Identified projects worth \$Ms in savings, pitched to SVPs/CTO, developed prototypes, and executed pilots with provider/payor groups
 - Managed collaborations with ML/NLP healthcare start-ups and academic labs; reviewed proposals, negotiated contracts, and oversaw successful project delivery
- Senior Data Scientist at **UPMC Enterprises** (09/2018–07/2019)
 - Developed a search engine for social determinants of health indicators in millions of clinical documents to identify high risk patients for intervention
- Internships at Microsoft Research (2013), Amazon (2009), Google (2008)

SELECTED PUBLICATIONS

- Jon Eskreis-Winkler, **Yubin Kim**, Andrew Stanton. 2023. [XWalk: Random Walk Based Candidate Retrieval for Product Search](#). In *Proceedings of eCom'23: ACM SIGIR Workshop on eCommerce*.

- Nithin Haridas, **Yubin Kim**. 2020. [Clustering Large-scale Diverse Electronic Medical Records to Aid Annotation for Generic Named Entity Recognition](#). In *Health Search and Data Mining Workshop*. 43–50.
- Zhuyun Dai, **Yubin Kim**, Jamie Callan. 2017. [Learning to Rank Resources](#). In *Proceedings of the 40th Annual ACM SIGIR Conference*. 837–840.
- **Yubin Kim**, Jamie Callan, Shane Culpepper, Alistair Moffat. [Efficient Distributed Selective Search](#). *Information Retrieval Journal* 20, 3 (2017), 221–252.

Full list available on [Google Scholar](#): search "Yubin Kim" on [scholar.google.com](#)

ACADEMIC SERVICE

- Organized Workshops, Panels, Symposia
 - Social media co-chair, SIGIR 2024
 - SIGIR Workshop on eCommerce, co-located with SIGIR 2023
 - CIKM Doctoral Symposium, CIKM 2022
 - SIRIP Panel on Future and Applications of Dense Retrieval, SIGIR 2022
 - WSDM Workshop on Health Search and Data Mining, co-located with WSDM 2020
- Peer Review
 - Editorial board of Foundations and Trends in Information Retrieval (FnTIR)
 - Program committee of SIGIR, CIKM, WSDM, TheWebConf
- Invited Talks and Panels
 - Panelist for SIGIR Symposium on IR in Practice (2023)
 - Panelist for "Data for Artificial Intelligence" hosted by *Value for Health CoLAB* (2021)
 - Panelist for "Top of Mind Exchange: AI" hosted by the *Center for Connected Medicine* (2020)
 - Invited talk titled "40,000 Note Types: EMRs in the Real World" presented at the *Artificial Intelligence for Data Discovery and Reuse Symposium* (2020), presented at Brown University (2019)